AI NOTES SERIES

Generative AI

AI that creates — how systems make new text, images, audio and video, what they help with, and where their risks and limits lie.

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These are introductory study notes, written in simple English for teachers, students and beginners. They come from my own learning and practical experience — creating and editing images with AI, and building websites with AI assistance. Explanations are simplified on purpose, so the main ideas are easy to follow and easy to revise.

What Generative AI Is

Generative AI is the part of artificial intelligence that **creates new content**. Instead of only answering with a label or a prediction, a generative system produces something that did not exist before — a paragraph, a picture, a voice recording, or a short video.

> **Plain definition:** Generative AI is AI that makes new text, images, audio or video by following patterns it learned from large numbers of examples.

Creating is different from classifying

Much of AI does not create anything. It sorts, labels or predicts. Seeing the difference makes the word "generative" clear.

- **Classifying AI** looks at something and gives it a label — for example, "this photo shows a cat" or "this message is spam". The output is a category.
- **Predicting AI** looks at past information and estimates what comes next — for example, tomorrow's weather or the price of a crop. The output is a forecast.
- **Generative AI** produces new material — for example, a story written from your idea, or a poster image made from your words. The output is content.

What it can create

- **Text** — essays, summaries, letters, lessons, questions and answers, and computer code.
- **Images** — pictures drawn from a written description, and edits to existing photos.
- **Audio** — spoken voice from written text, and music or sound made to order.
- **Video** — short moving clips made from text, or from a still image set in motion.

A generative system does not copy one example and hand it back. It mixes patterns from many examples into something new. That is why two people can ask for similar things and receive different results — and why the same person can ask twice and get two different answers.

How It Works, Simply

Behind the friendly screens, the basic story is the same for text, image, audio and video systems. A model studies examples, learns the patterns inside them, and then builds new material in the same patterns.

Very many examples

↓

The model learns the patterns

↓

You give a prompt — your instruction

↓

New content in the same patterns

Learning patterns

- During **training**, the model is shown very large numbers of examples — sentences, photographs, recordings — and adjusts itself until it can predict what usually comes next or what usually belongs together.
- For text, this means learning which words follow which other words. For images, it means learning how shapes, colours and light usually fit together.
- The trained result is called the **model**. When you use a tool, you are using the model, not the original examples.

The honest point: plausible is not the same as true

A generative model is built to produce output that **looks plausible** — that fits the patterns well. It is not built to check facts. A sentence can be smooth, confident and completely wrong. A picture can look like a real photograph of something that never happened.

> **Remember:** Generative AI optimises for what looks right, not for what is true. Checking facts, names, dates and numbers is always the human reader's job.

This is not a reason to avoid these tools. It is a reason to use them the way a careful student uses a first draft — as a starting point to check, correct and improve.

The Forms

Generative AI is not one tool. It appears in several forms, and many modern tools combine more than one form in a single product.

Text generation

The best-known form. You write a request, and the system writes back — a draft letter, a lesson explanation, a summary, practice questions for a class, or help with computer code.

Image generation and editing

You describe a picture in words, and the system draws it. The same tools can edit an existing photo — changing a background, removing an unwanted object, or improving light and colour. In my own practice this is the form I use most: creating teaching visuals and editing photographs, always checking that the result still looks honest.

Video and image-to-video

Some systems make short video clips from a written description. Others start from a still image and add movement — a portrait that smiles, or a landscape where water flows. Results are usually short, and small errors in hands, faces and movement are still common.

Speech and audio generation

Written text can be turned into natural-sounding speech in many languages, and systems can also generate music and sound effects. This helps greatly with listening material, and for people who find reading difficult.

Multimodal creation

Multimodal tools work across kinds of content at once — you can show a photo and ask questions about it in words, or give text and receive an image with a spoken explanation.

AI-assisted website creation

Generative tools can draft the words, layout and code of a website, which a person then checks, corrects and publishes. I have used this approach for real websites: the AI produces a fast first version, and the human decides what is accurate and suitable.

Creative automation

Because generation is fast, one idea can become many drafts in minutes — several versions of a poster, a headline, or an illustration style. The human chooses and improves the best one instead of starting from a blank page.

What It Helps With

Used with care, generative AI is a practical helper — especially for teachers, students and small teams who have more ideas than time, money or specialist staff.

Drafting

- A first draft of a letter, notice, lesson plan or report in minutes, ready for a person to correct and finish.
- The same material rewritten at an easier level for younger students, or in simpler English for beginners.
- Ideas when the page is blank — headings, examples and questions to react to, instead of nothing at all.

Visual material for teaching

In many schools there is no photographer, no artist and no budget for printed charts. A teacher can generate a clear picture of a volcano, a historical scene or a science process for the classroom wall or a worksheet. The picture must still be checked — labels and details in generated images can be wrong — but a checked generated image is far better than no visual at all.

Prototyping ideas

- See what a poster, a logo idea or a web page could look like before spending money on it.
- Try several styles quickly, then invest real effort only in the direction that works.
- Show others a visible draft instead of only describing an idea in words.

Accessibility formats

The same content can be offered in the form each person needs: a written lesson as text, the same lesson read aloud as speech for someone who cannot see well or reads with difficulty, and key ideas as a simple image for a young learner. Generative AI makes moving between these formats fast and cheap.

> **The pattern in every case:** the AI produces a fast first version, and a person checks it, corrects it and takes responsibility for the final result.

The Risks

The same abilities that make generative AI useful also create real risks. Knowing them clearly is part of using the technology well.

Misinformation at scale

False text, images and audio used to take time and skill to produce. Now they can be made quickly and in large amounts. A fake picture of an event, or a false news story written in perfect language, can spread before anyone checks it. The defence is habit: pause, check the source, and compare with a trusted source before believing or sharing.

Copyright questions

- Models are trained on material made by other people — books, articles, photographs and artwork. Whether that training use is always fair is still debated by courts, governments and creators.
- Outputs can sometimes come out very close to an existing work. Who owns a generated picture or text, and when it copies too much, are questions without one settled answer everywhere.
- The careful practice is simple: do not present generated work as if you created every part of it alone, and do not use generated copies of a living artist's or writer's work as your own.

Identity misuse

Synthetic images, video and voice can be made to look or sound like a real person saying or doing something they never said or did. This misuse is often called a **deepfake** when it imitates a real person. It can damage reputations and trust, and it is wrong even when it is meant as a joke.

> **Two firm rules:** Label generated content as generated, so nobody is misled about where it came from. And never use generated images, video or voice to impersonate anyone — not a public figure, not a colleague, not a friend.

Your own protection

- Be careful with photos and voice recordings of yourself and your family that you share with tools and websites.
- Agree a family check — for example, a code word — for any surprising voice message asking for money or urgent help.
- When a picture or clip looks shocking, treat that feeling as a signal to verify it, not to forward it.

Honest Limits

Generative AI is powerful, but a clear view of its limits stops both disappointment and over-trust.

Output quality varies

- The same request can give an excellent result once and a poor result the next time. Generation involves chance, so results are not perfectly repeatable.
- Small details are the weakest point — fingers in a picture, spelling on a sign inside an image, numbers and dates in a text.
- Topics and languages with less training material, including many local languages and local subjects, usually get weaker results.

Styles move toward the average

A model learns from very many examples, so its natural pull is toward what is most common in them — the average style, the average answer. Left alone, generated work can feel smooth but ordinary, and different tools can start to sound and look alike. Original ideas, local detail and a personal voice have to come from the human.

Human editing is the difference

In my own work with generated images and AI-assisted websites, the pattern is always the same: the raw generation is only the middle of the job. What makes the final piece good is human work around it — choosing the best version, correcting errors, adjusting details, checking facts, and deciding what is suitable to publish.

> **Honest summary:** Raw generation is fast and cheap. Good work is still human work — generation just gives that work a better starting point.

Other limits to keep in mind

- A model only knows patterns from its training; it does not know today's events unless the tool is connected to fresh information.
- It cannot take responsibility. If a generated letter, lesson or picture is published, the person who published it is responsible for it.
- It does not replace learning. A student who only collects generated answers has drafts, but not knowledge.

Words to Know

Generative AI — AI that creates new content, such as text, images, audio or video, by following patterns learned from large numbers of examples.

Prompt — the instruction or request you give to a generative system: the words, and sometimes the image, that tell it what to make.

Model — the trained result of machine learning: the stored patterns a system uses to generate new content or answer new requests.

Training data — the examples a model learned from during training, such as text, photographs, artwork and recordings made by people.

Multimodal — able to work with more than one kind of input or output together, for example text with images, or speech with video.

Deepfake — synthetic media made to imitate a real person, such as a fake video or voice recording of someone saying or doing something they never did.

Copyright — the legal right of a creator to control how their work is copied and used; with AI it raises questions about both training material and generated outputs.

Label — a clear note attached to generated content saying that AI made it, so readers and viewers are not misled about where it came from.

Quick Revision

- Generative AI creates new content — text, images, audio and video — instead of only classifying or predicting.
- The model learns patterns from very large numbers of examples, then produces new combinations in the same patterns.
- It optimises for what looks plausible, not for what is true. Always check facts yourself.
- Its forms include text, image generation and editing, video, speech and audio, multimodal creation, AI-assisted websites and creative automation.
- It helps with drafting, teaching visuals where no photographer exists, prototyping ideas, and accessibility formats.
- Its risks include misinformation at scale, copyright questions, and identity misuse through synthetic images and voices.
- Label generated content as generated, and never use it to impersonate anyone.
- Output quality varies, and styles move toward the average of the training data.
- Human editing is the difference between raw generation and good work.

Closing Note

These notes are an introduction to generative AI. Explanations are simplified on purpose, so a beginner can build a clear picture first and add detail later. They are study notes from my own learning and practical experience — creating and editing images, and building websites with AI assistance — shared for students, teachers and anyone starting their journey with artificial intelligence.

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